

KWA-THEMA SKILL SCHOOL

NS-TECH WORKBOOK

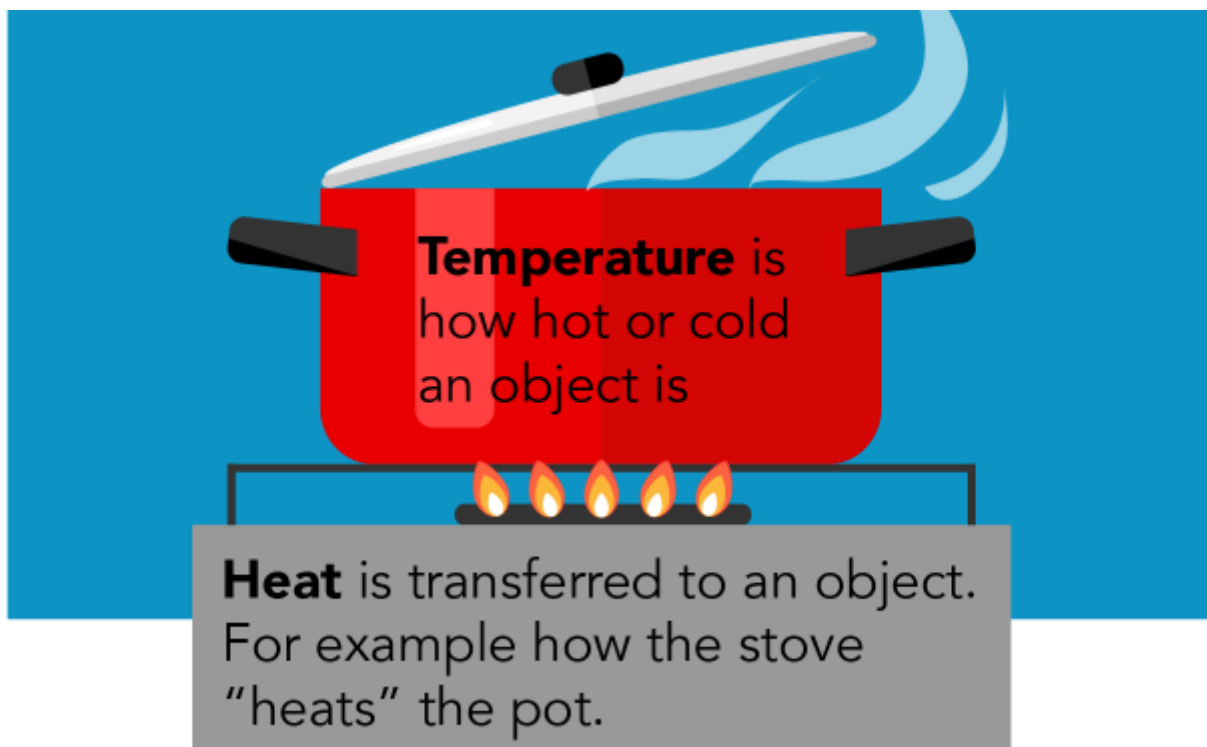
YEAR 3 TERM 3



HEAT



TEMPERATURE



NAME: _____

YEAR: _____

NS YEAR3 TERM 3

cuemath
THE MATH EXPERT

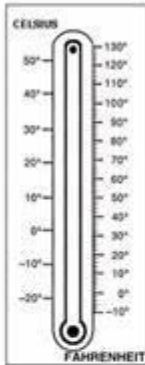
A collection of 15 different types of thermometers. The items include: a long horizontal analog thermometer with a red dot at 37; a purple digital thermometer; a green digital thermometer; a blue and white digital thermometer; a yellow digital thermometer; a blue digital thermometer with a cord; a yellow and black infrared thermometer; a pink and white infrared thermometer; a circular analog thermometer with a red needle; a large yellow and black infrared thermometer; a pink digital thermometer; a black and green digital thermometer; a circular analog thermometer with a red needle; a large vertical thermometer with dual scales (C and F); and a large orange thermometer with a dual scale and a small circular gauge at the bottom.

HEAT, TEMPERATURE, CELCIOUS SCALE, BOILING POINT, MELTING PONT, CONDUCTOR, INSULATOR, THEMOMETER

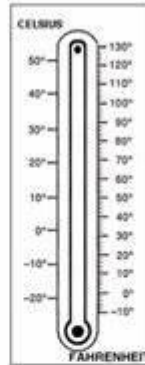
Name: _____

The Right Temperature

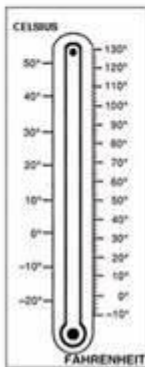
Directions: Look at the pictures below. Does it show temperatures that would be hot, warm, cold, or freezing? Write a temperature and fill in the thermometer for each picture. Write how the temperature feels below each picture.



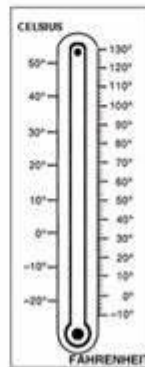
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Graphics (c) <http://www.schenectady.k12.ny.us>

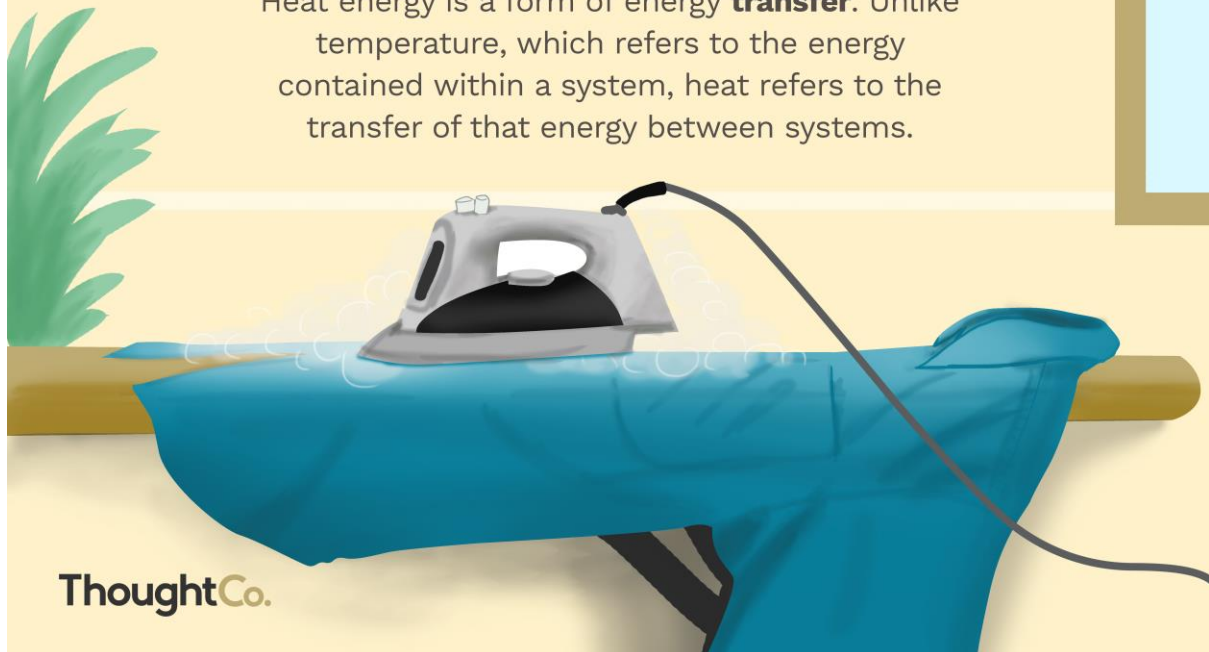
©www.HaveFunTeaching.com

answers

NOTE: THE LEARNER HAS TO UNDERSTAND DEFINATION AND MEANING OF THE ABOVE CONCEPTS ONLY THEN THEY CAN MATCH THE CONCEPT WITH THE DESCRIPTION

Heat Energy Definition

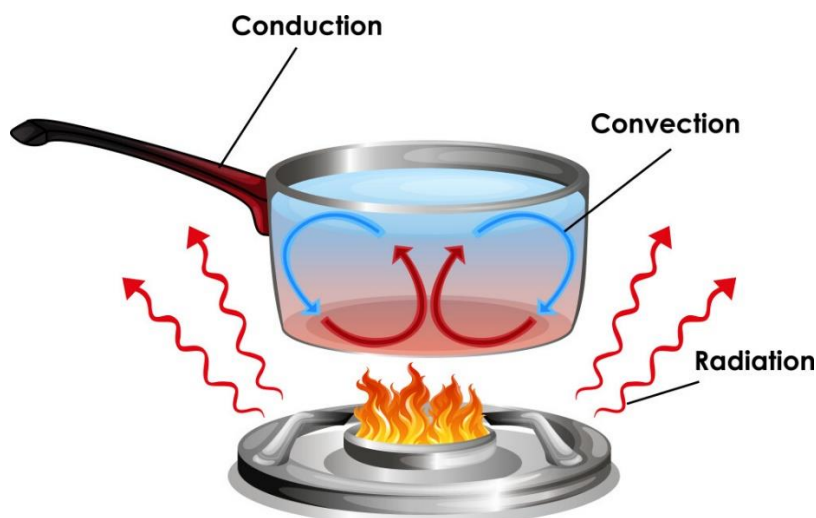
Heat energy is a form of energy **transfer**. Unlike temperature, which refers to the energy contained within a system, heat refers to the transfer of that energy between systems.



HEAT: Is the transfer of energy from one body to another as a result of a differences in temperature. When two bodies of differing temperature(warm/cold) come together energy is transmitted, and heat is transferred (--) from (hotter) to (colder)

3WAYS HOW ENERGY CAN BE TRANSFERRED

1.Conduction 2. Convection 3. Radiation



Heat Transfer

MORE EXPLANATORY PICTURES

Worksheet: Methods of Heat Transfer (conduction, convection, and radiation)

Define conduction: transfer of heat by direct contact between materials

Define convection: transfer of heat by movement of large volumes of fluids moving to balance average KE (temperature)... hot rise, cool falls

Define radiation: transfer of heat not requiring a medium to move it

Identify the method of heat transfer that takes place in each illustration. Some illustrations may show more than one form of heat transfer.

1. 
Radiation
2. 
convection
3. 
conduction
4. 
Radiation
5. 
radiation
6. 
radiation
7. 
Radiation
8. 
radiation
9. 
convection
10. 
Radiation, conduction
11. 
conduction
12. 
convection

ACTIVITY:BASED ON CONVECTION,CONDUCTION,RADIATION

Name: _____ Period: _____ Date: _____

Convection, Conduction, and Radiation

Define conduction:

Define convection:

Define radiation:

Identify the method of heat transfer that takes place in each illustration. Some illustrations may show more than one form of heat transfer.

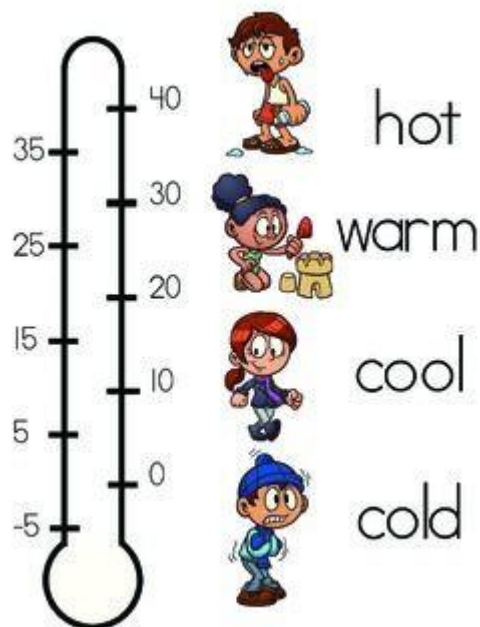
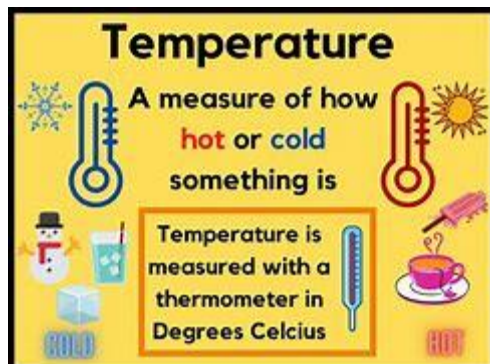


Define: conduction	Convection:	Radiation:
1.	2.	3.
4.	5.	6.
7.	8.	9

10.

11.

12.



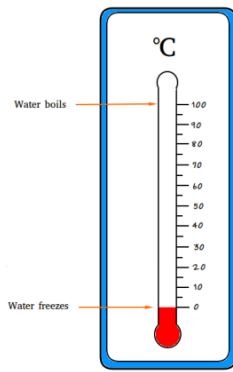
Temperature: What's HOT & COLD

HOT	COLD

Created by Beth at www.livinglifeintentionally.blogspot.com

LOOK AT THE ABOVE PICS THEN ANSWER

Hot		cold	



BOILING POINT

Boiling Point

The boiling point is the temperature at which a liquid changes into a vapor (boils).

Boiling Point of Water

at sea level 100 °C (212 °F)		at 6250 feet 93.4 °C (200.1 °F)
---	--	--

At the boiling point, the vapor pressure of the liquid is the same as the vapor pressure of air (or whatever is outside the liquid).

sciencenotes.org

BOILING POINT

- The temperature at which a liquid boils.
- The temperature at which a liquid changes into a gas.
- The boiling point is dependent on pressure.



FAST FACT!

The boiling point of water at sea level is 100° Celsius. On top of Mount Everest the boiling point of water is 69° Celsius. This happens because of the difference in pressure.

10

THE MELTING POINT



Melting point is the temperature at which a solid substance changes its state to become a liquid.

It is the temperature at which the thermal energy (heat) is sufficient to break the intermolecular forces that hold the particles of a solid together, allowing them to move freely and take on the properties of a liquid.

The **melting point** is a physical property of a substance and can be used to identify and characterize it.

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ACTIVITY :YOU NEED TO CONNECT THINGS THAT CAN MELT/FEEZE

Freezing and Melting - Things you can melt

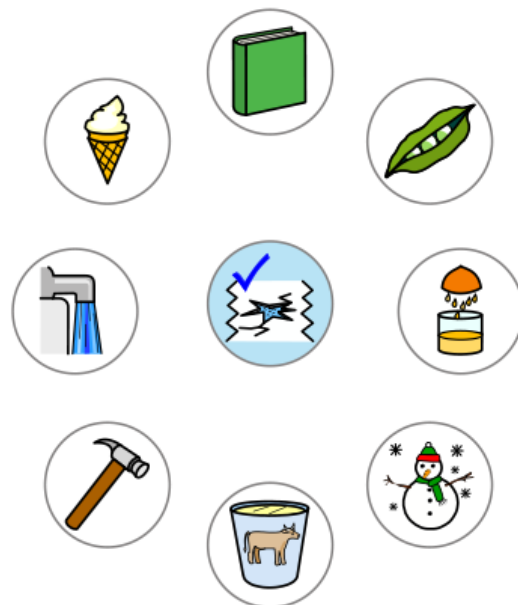
Connect the things that can melt



Widgit Symbols © Widgit Software 2002-2017

Freezing and Melting - Things you can freeze

Connect the things that can freeze



Widgit Symbols © Widgit Software 2002-2017

Heating Materials

Materials	What happens when we heat them?	Can we change them back again?
metal 		
cheese 		
eggs 		
wood 		
chocolate 		
bread 		
wax 		
milk 		
clay 		

Hamilton Trust 2007 Y2 - 5c - Changing materials - Session D

The table below shows the melting point and boiling points of 4 substances, E, F, G and H.

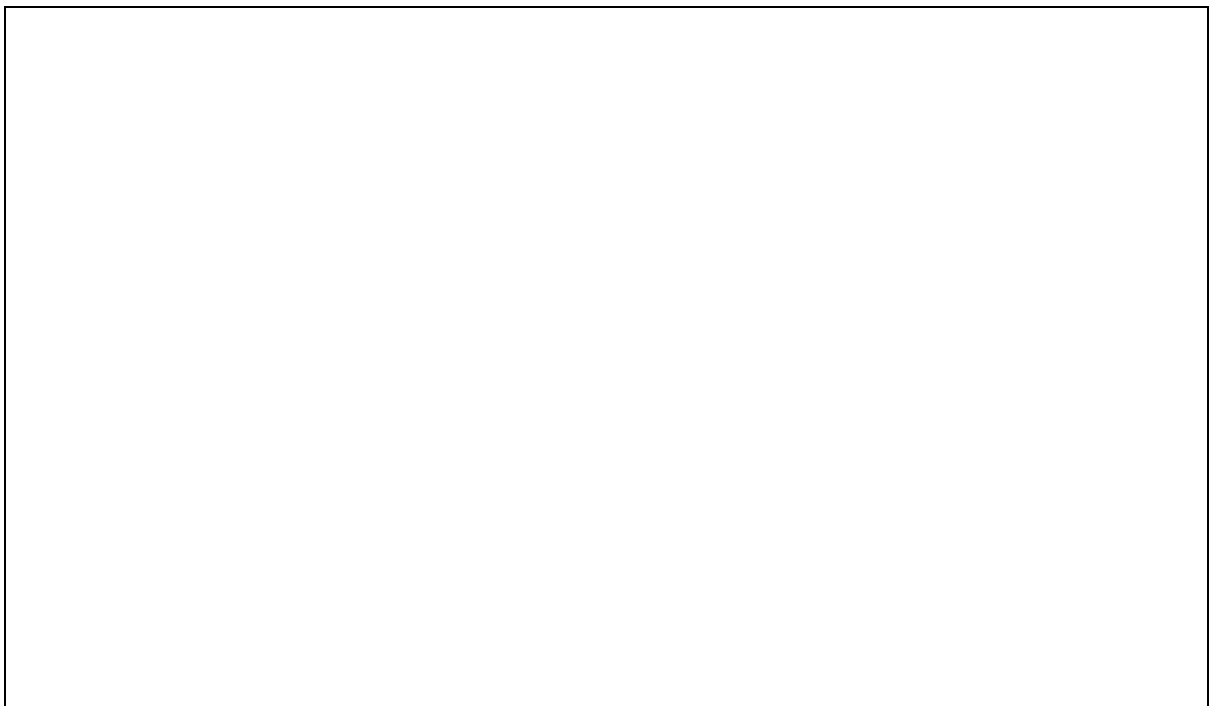
Substances	Melting point (°C)	Boiling point (°C)
E	70	99
F	23	67
G	47	120
H	15	36

At which temperature, **will only two** substances be found in the liquid state?

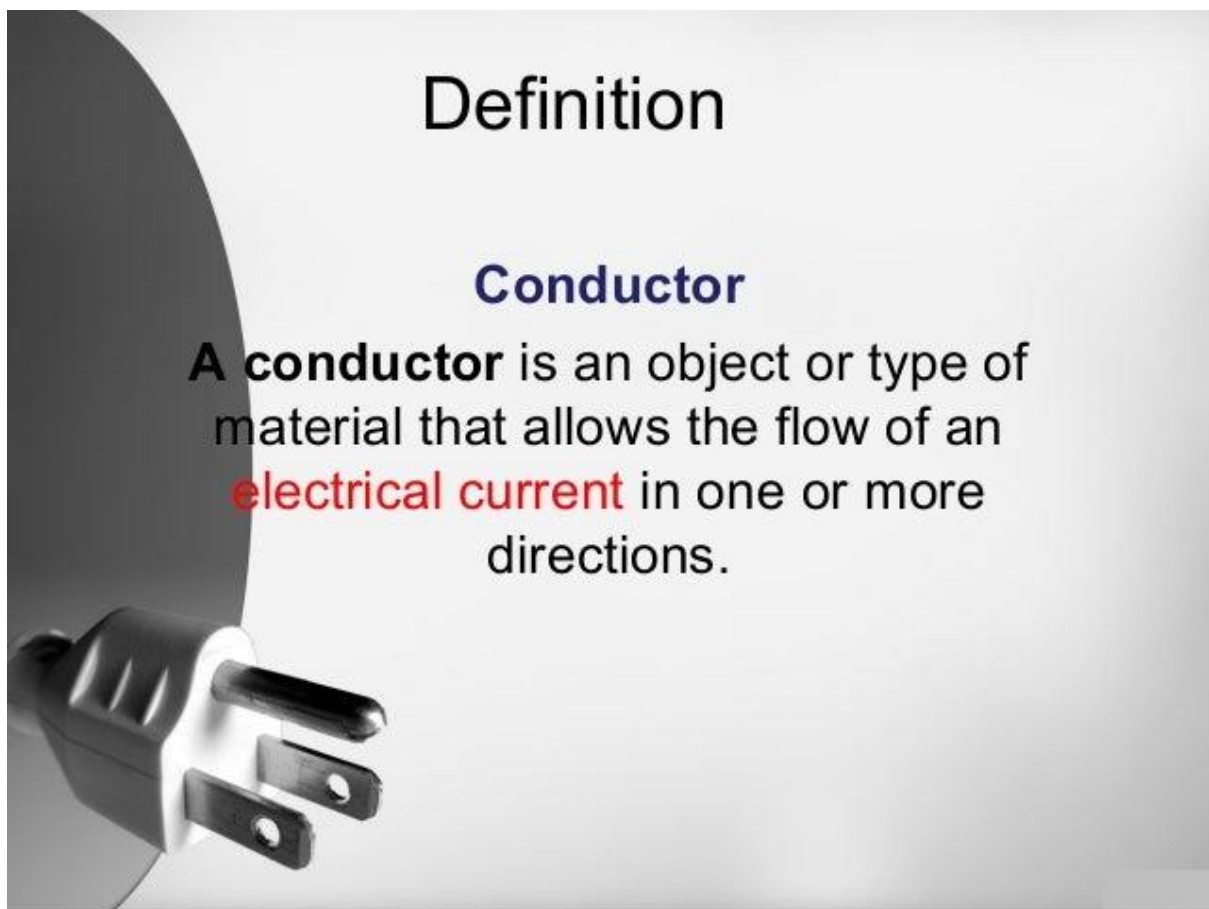
- (1) 22 °C
- (2) 45 °C
- (3) 78 °C
- (4) 100 °C

()

ACTIVITY: USE THE ABOVE INFORMATION TO DESIGN 2 GRAPHS CHOOSE BETWEEN A PIE/BAR /LINE GRAPH (COLOUR) IT USE THE SPACE PROVIDED (10)

A large, empty rectangular box with a thin black border, intended for drawing a graph. It occupies the upper half of the page below the instructions.A second large, empty rectangular box with a thin black border, identical to the one above, intended for drawing a second graph. It occupies the lower half of the page.

CONDUCTOR

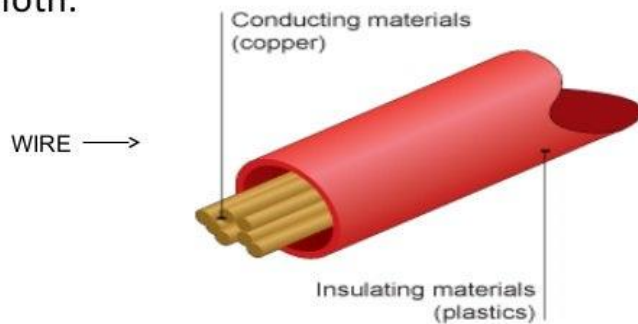


INSULATOR

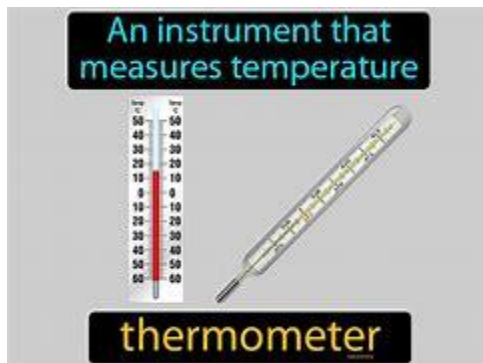
Are non -conducting material that do not easily allow current to pass through them

Insulators

- **Insulators:** They don't easily allow the flow of electric charges through them.
- Ex: glass, plastic, certain rubber materials, cloth.



THERMOMETER



ACTIVITY: LEARNER HAS TO MATCH THE CONCEPT WITH THE DESCRIPTION

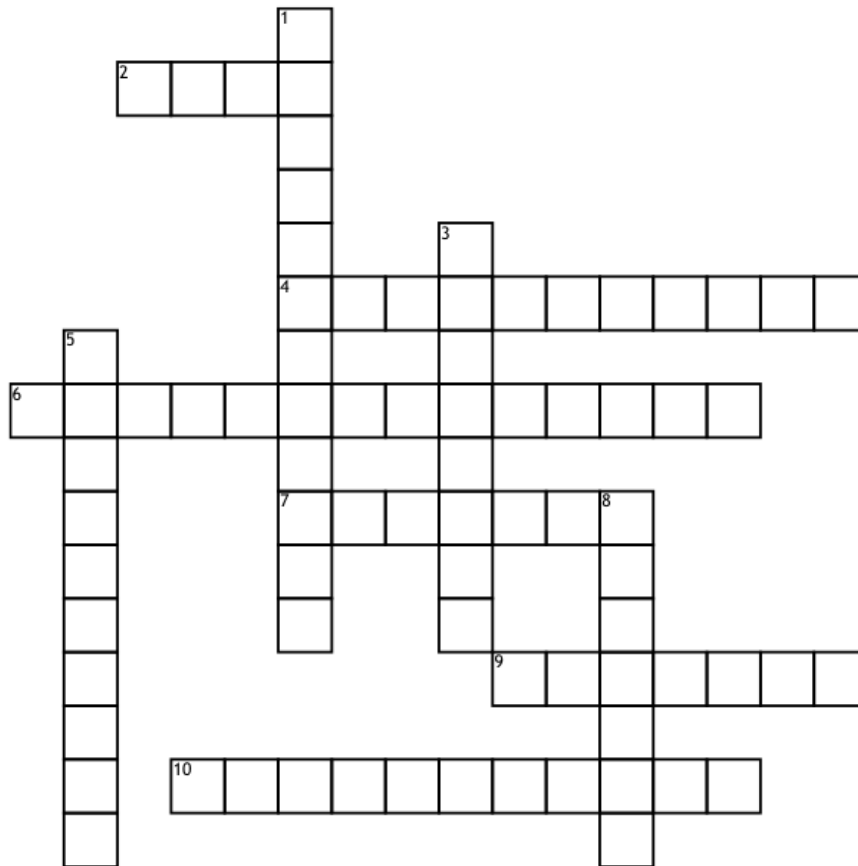
HEAT, TEMPERATURE, CELCIOUS SCALE, BOILING POINT, MELTING PONT, CONDUCTOR, INSULATOR, THEMOMETER, HEAT ENERGY

DESCRIPTION	CONCEPTS
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1.An instrument that measure temperature	
2.A conductor is an object or type of material that allows the flow of An electrical current in one or more direction	
3.This is the temperature at which solid substance changes its state to become a liquid	
4.They don't allow the flow of electric charges through them	
5.Is the temperature at which a liquid change into a vapor	
6.Heat nergy is the form of energy transfer refers to energy contained within a system	
7.Is the transfer of energy from one body to another as a result of a difference in temperature	

Name: _____ Date: _____

Temperature Crossword



Across

- 2. 37 degrees celsius is our ... temperature
- 4. A tool used to measure temperature
- 6. The unit of measurement we use in SA
- 7. What we call the highest temperature
- 9. 100 degrees celsius is ... point

- 10. A measure of the warmth or coldness of an object or substance

Down

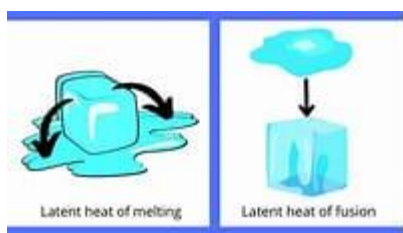
- 1. If we experience very cold temperatures we can get ...
- 3. 0 degrees celsius is ... point
- 5. If we experience very hot temperatures we can become...
- 8. What we call the lowest temperature

(10)

PROCESS OF BOILING AND MELTING POINT

About 694 000 results

1.



Melting and boiling are processes in which these noncovalent attractive interactions are disrupted. The stronger the intermolecular attractive forces, the more energy that is required to break them apart. That energy is in the form of heat, e.g. a higher temperature must be reached for the melting or boiling to occur.

NB *FROM THE SMARTBOARD: DEMONSTRATION OF HOW TO READ A THERMOMETER MUST BE VIEWED

NB* DEMONSTRATION ON HOW HEATING AND COOLING CAN CHNGE MATERIALS

ACTIVITY: INVESTIGATION

MATERIAL AND APARATUS	METHOD OF THEIR OWN IS EXPLAINED BY THE LEARNER TO NVESTIGATE THE COOLING AND HEATING PROCESS
Glass beaker (2) Bunsen burner/one plate stove Tripod with gauze Tap water Thermometer Funnel Ice blocks/freezer	WHAT NEEDS TO HAPPEN WATER TO MAKE IT FREEZE?

	WHAT NEEDS TO HAPPEN TO MAKE IT BOIL ?

	(10)

2. State the 3 stages of matter and give examples

_____	_____	_____
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(6)

3.Can you indicate How do you measure the temperature for the above experiment

FOR THE HOT PROJECT	FOR THE COLD
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(4)

4.What is the temperature for

HOTTER STATE _____	FREEZING STATE _____
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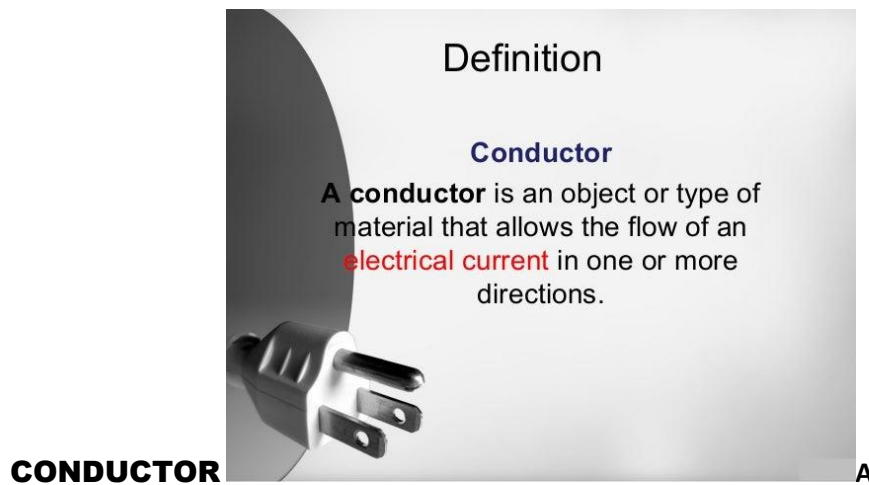
(2)

BEFORE YOU COMPLETE THIS ACTIVITY LET US OBSEVE YOUR INVESTIGATION ON HEATING AND COOLING PROJECT AND WRITE YOUR OBSERVATION IN THIS TABLE

ELAPSED TIME (MINUTES) ON 1.HEATING 2.COOLING	TEMPERATURE OF WATER
1.	
2.	

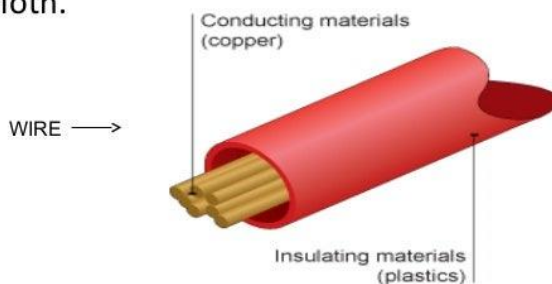
(4)

INVESTIGATION ON CONDUCTORS AND INSULATORS (REMEMBER THIS PICTURE



Insulators

- **Insulators: They don't easily allow** the flow of electric charges through them.
- Ex: glass, plastic, certain rubber materials, cloth.



NB(GOOGLE)

ACTIVITY: INVESTIGATING CONDUCTORS AND INSULATORS

HOW WILL YOU KNOW THESE MATERIAL CONDUCT ELETRICITY?

CONDUCTORS AND INSULATORS



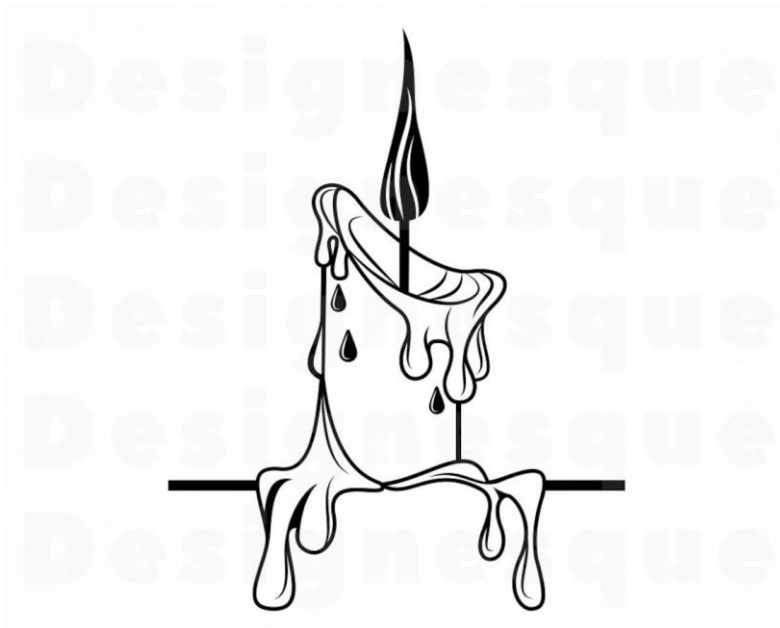
CONDUCTORS (explain and give examples)	INSULATORS (explain and give examples)
1.	1.
2.	2.
3.	3.

NB (OUR LEARNERS CAN NOT CONNECT ELECTRIC WIRES THEY CAN WATCH VIDEO ON SMART BOARD THEN ANSWER

VIDEO VIEWED*****

1.What type of material did not light up ?

2.What does that mean?



3.Write a short story what is happening with this candle

4.What was the mistake here

5. Do you think its wise to put a candle like this and what will happen when it reaches the final stage